

Concrete grinding and polishing

How to match disc speed, diamond segments, head pressure and travel speed to the floor hardness — for a flat floor and a clean, lasting gloss

Draft for review by Xingyi engineering

This guide was prepared by Jarek Walicki (CleanVisions) for operators and customers of Xingyi planetary grinders. Before it is used with customers, we would like Xingyi engineers to check it, correct it and confirm the numbers for Xingyi machines. Questions for the review are on the last page.

The rule everything starts with

Gloss is not proof of quality. Gloss proves that someone polished. Flatness proves that someone ground well.

Flatness is created in steps 1 and 2 — with metal and hybrid tools. Resin cannot fix it later. If the slab is not levelled at the beginning, no amount of polishing will repair it.

Five things that must match each other

A machine does not grind well on its own. The result depends on five elements working together. Changing one changes how the others work:

- **Concrete hardness** — decides the segment bond.
- **Segment bond and grit** — decide whether the diamond cuts into the surface or just slides.
- **Disc speed (RPM)** — calmer for coarse grinding, higher for polishing.
- **Head pressure (machine weight, counterweight position)** — highest for coarse grinding, lower for polishing.
- **Travel speed** — decides how long the tool works on each spot. Too fast = not flat.

Step 0 — read the concrete before you fit the segments

The concrete class from the design (e.g. C25/30) describes compressive strength, not surface hardness. It is only a guide. Tool selection is decided by what you check on site:

- **Scratch test** in several places — surface hardness often varies within one hall (near gates and in traffic lanes it is often harder than further inside).
- **Test run on 1–2 m²** — watch and listen: does the segment remove material and produce slurry or dust, or does it slide, heat up and squeal?
- **What is on the surface** — dry-shake hardener, old coating, adhesive, densifier. All of these change the hardness and the tool choice.
- **Flatness** — use a straightedge or side light to find waves, steps and holes.

Choosing the segment bond — the inverse rule

A diamond only works while the metal bond wears away and exposes new grains. That is why the bond is chosen **opposite** to the concrete hardness:

Concrete	Segment bond	Why	What happens if it is wrong
Hard — power-trowelled, with hardener, densified	soft	Hard concrete must wear the bond so that new diamonds are exposed.	Hard bond: diamond glazing — the segment slides, heats up, removes nothing, the machine may overload.
Medium	medium	Balance between productivity and wear.	—
Soft — sandy, abrasive, screed	hard	Loose sand quickly eats a soft bond.	Soft bond: the segment disappears fast , and loose grains can cause deep scratches.

Quick check while working: the machine slides and the concrete heats up without result → switch to a softer bond. The segment wears very fast → switch to a harder bond.

Machine settings step by step

The finer the grit, the **lower the pressure** and the **higher the RPM**. Travel speed must be **steady and calm** at every step — it does not simply increase with the grit.

Step and tool	Goal	RPM	Pressure / counterweight	Travel speed	When you may move on
1. Coarse metal (e.g. 16–40#)	Remove coatings and laitance, open the concrete, take the high spots down to the level of the low spots .	low to medium	highest (counterweight front)	slow — the segment needs time to work	The whole surface is evenly opened, no "islands" of old surface. Holes and cracks filled.
2. Finer metal (e.g. 60–120#)	Remove scratches from step 1, finish levelling.	medium	high to medium	slow, steady, overlapping	After washing: not a single scratch from step 1 under side light.
3. Hybrid	Remove metal scratches without leaving its own. Transition to resin.	medium	medium (counterweight middle)	steady, overlapping	After washing: uniform matte surface, no metal scratches.
4. Resin 50–200#	Still grinds — removes the hybrid pattern. Grouting and densifying usually happen here.	medium to higher	medium	steady, overlapping	Uniform fine matte, no scratches. Densifier only when steps 1–2 are finished.
5. Resin 400# and up	Polishes — closes and smooths the surface, builds gloss and reflection clarity.	higher	lower (counterweight rear)	steady, no rush	Each step gives a clearly sharper reflection. No burn marks or streaks.

The grits in the table are an example of a typical sequence — the exact set depends on the concrete and the tools. Never skip more than one grit at a time.

Suggested settings — a starting point

This is a suggestion, not a fixed scheme

The values below are a **starting point** from field practice with heavy planetary grinders. Start in the middle of the range, do a test run on 1–2 m² and adjust: watch, listen, wash and check under side light. If the machine manual gives different values — **the manufacturer's manual wins**.

For Xingyi engineering: please confirm or correct these ranges for Xingyi machines.

Step and tool	Bond	Disc speed (RPM)	Pressure	Travel speed	Watch out for
Coarse metal / PCD 16–30#	opposite to concrete: hard concrete → soft, soft concrete → hard	350–800 hard concrete: approx. 700–900	maximum counterweight front (100%)	3–8 m/min	Slow and thorough. This is where flatness is made. RPM too high = glazing and overload.
Metal 30–70#	as above	800–1200	high front or middle (70–100%)	6–12 m/min	Removes coarse metal scratches and finishes levelling. Wash before the next step.
Metal 70–120#	as above	1000–1400	medium counterweight middle (approx. 70%)	8–14 m/min	After washing, no scratch from earlier steps may remain.
Hybrid 50–200#	—	1200–1500	medium counterweight middle (approx. 70%)	10–15 m/min	Removes metal scratches without leaving its own. Do not skip this step.
Resin 50–200#	—	1400–1650	medium counterweight middle (approx. 70%)	12–18 m/min	Still grinding. After this step: grouting and densifier — only when the floor is flat.
Resin 400–800#	—	1650–1850	lower middle or rear (65–70%)	12–20 m/min	Polishing. Too much pressure heats the resin and leaves streaks.
Resin 1500–3000#	—	1800–1950	lowest counterweight rear (approx. 65%)	12–20 m/min	Steady, no rush. Do not stop in one place — you will burn the surface.

How to read these numbers

- **RPM:** some manufacturers display motor speed, others disc speed. Check in the manual what your display shows.
- **Pressure:** the percentage means the share of the machine's maximum head pressure. Example: a machine that gives approx. 380 kg on the heads in the front position gives approx. 270 kg in the middle and approx. 250 kg in the rear. The exact values are in your machine's manual.
- **Travel speed:** 6 m/min is approx. 10 cm per second, 12 m/min approx. 20 cm per second, 18 m/min approx. 30 cm per second. **The machine's maximum travel speed is a limit, not a working setting** — do not run at maximum even with the finest resin.
- **Adjustment rule:** change one thing at a time. First the bond (when the segment slides or disappears), then pressure, then RPM, and finally travel speed.

Six rules that decide flatness and gloss

1. Step 1 takes as long as the floor needs.

Sometimes short, sometimes very long. Do not move to step 2 until the slab is level, holes are filled, and cracks and joints are repaired.

2. Do not speed up to show progress.

Defects most often appear when the customer wants a fast handover and the operator speeds up. Fast travel = short contact = waves and under-ground areas remain.

3. Wash the floor between steps.

Dust left after vacuuming hides fine scratches and unevenness. They only show up when polishing after washing — then you may have to go back as far as the metals. Wash, dry, check under side light, and only then change grit. Also clean the underside of the machine and the tool holders — one grain from a coarse grit will scratch a fine step.

4. Work with overlap, in two directions.

Each pass overlaps the previous one (typically about 1/3 of the width). On coarse grits, cross passes (lengthwise, then crosswise) help level the slab.

5. Densifier only after checking.

After densifying, the surface is harder. If scratches or unevenness show up afterwards, going back to metals is much more difficult. Check and level first, densify after.

6. Resin does not close pores and does not level.

Pores are closed by grouting, the surface is strengthened by densifier. Resin smooths and polishes. Too much pressure, too little water or standing still overheats the resin — streaks and burn marks remain.

Quick diagnosis — symptom, cause, what to change

Symptom	Most common cause	What to change
Machine slides, squeals, heats up, removes nothing	Diamond glazing — bond too hard for the concrete, sometimes RPM too high	Softer bond. As a quick fix, dress the segments on an abrasive material (dressing block, sandstone) to expose them.
Segments wear very fast	Bond too soft on soft, sandy concrete	Harder bond.
Overload protection tripped	Too much pressure on coarse grit, glazing, voltage too low, cable too long or too thin	Disconnect power, wait for it to cool down as per the manual. Remove the cause, not only the effect.
Circular scratches visible after washing	Skipped grit, step too short, dirty tools, no washing between steps	Go back to the grit that did not remove the scratches. Wash the floor and the machine between steps.
Light, matte patches in the gloss	The slab was not levelled in steps 1–2. The resin "copies" the wave instead of removing it	Cannot be fixed after polishing. Requires going back to metal in those areas.
Dark, burnt areas or streaks	Overheated resin: too much pressure, too little water, too long in one place	Less pressure, steady travel, do not stop in one place. Check the condition of the tools.
Small holes and pits in the matrix	Grouting skipped after opening the concrete	Grout before polishing. Every pinhole collects dirt.

How to check the result — not only by eye

Gloss alone (GU reading) is not enough to judge the work. A floor can shine and still not be flat. The simplest check: look at the reflection of a lamp or a window in the floor.

What you see in the reflection	What it means
Sharp outline of the lamp	Full grit sequence, surface properly closed.
Bright but blurred outline, haze	Skipped intermediate grits, or a coating on an uneven substrate.
Reflection waves as you look into the distance	The slab is not flat — this is unevenness, not a gloss problem.
Reflection breaks up in patches	Low spots not polished — levelling missing in steps 1–2.

- **For handover**, measure: gloss (60°), distinctness of image (DOI) and roughness — at several points, always with the same instrument.
- **Slip resistance**: polishing changes the friction of the floor. When finished, check slip resistance, especially at entrances and where the floor can get wet.

Health and safety — always

Concrete dust contains crystalline silica. Do not grind dry without a working industrial dust extractor matched to the machine and a clean filter. A clogged filter also overheats the segments.

Power supply: a cable that is too long or too thin and voltage drops overload the drive. Cable cross-section and minimum voltage — as per the machine manual.

Questions for Xingyi engineering

To turn this draft into a guide we can give to Xingyi customers, we need your confirmation or correction:

1. Are the RPM ranges on page 3 correct for Xingyi planetary grinders? Does the display show **disc speed or motor speed**?
2. What is the head pressure in kg for each counterweight position on your main models?
3. What travel speed do you recommend for metal, hybrid and resin steps? Is there a maximum you advise against for polishing?
4. Do you have factory recommendations for segment bonds (hard / medium / soft) and your bond colour codes by concrete hardness?
5. Minimum working voltage, recommended cable cross-section by cable length, and cooling time after overload protection trips?
6. Anything in this guide you disagree with, or would add from your experience?

Reviewed by Xingyi

Name: _____ Date: _____

Approved [] Approved with corrections [] Not approved []

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